

The impact of self-disclosing chatbots for academic stress assessment on student self-reflection

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Abstract

Academic stress profoundly impacts students' well-being and academic performance, highlighting the need for effective tools that foster self-reflection to enable meaningful academic stress assessment. This study explores the effectiveness of incorporating self-disclosure in chatbots to encourage students' self-reflection on their academic stress. Two chatbots were developed: non-self-disclosure chatbot (NSD) and self-disclosure chatbot (SD). An experiment was conducted with 20 university students who interacted with their assigned chatbot. Analysis of the interview responses revealed that the SD chatbot fostered deeper self-reflection, aiding in clear identification of stress factors and planning for future actions. In contrast, the NSD chatbot failed to sufficiently elicit self-reflection due to the superficial conversations. Additionally, students expressed expectations for educational uses of self-disclosure chatbots and concerns about privacy and data security. Based on these findings, we propose design recommendations for developing chatbots that support meaningful academic stress assessments by fostering self-reflection.

Keyword

Chatbot; Academic stress assessment; Self-disclosure; Student; Self-reflection

1. Introduction

Academic stress is a critical factor influencing students' well-being and academic performance in higher education [1]. Effective management of academic stress requires a comprehensive understanding of specific stressors, symptoms, and coping strategies, which can be achieved through in-depth academic stress assessments [2]. Central

to these assessments is self-reflection—a process that allows students to examine their own experiences and responses to stress more deeply. By engaging in self-reflection, students can identify and articulate the unique source of their stress, how it emerges as symptoms, and what coping strategies they rely on, leading to more precise and meaningful stress assessments [3].

Current approaches for assessing academic stress, such as self-report questionnaires and professional counseling, are widely used for their advantages but still face significant limitations. Self-report questionnaires like the SISCO Inventory of Academic Stress (SISCO-AS) [4] are commonly used due to their accessibility and simplicity. However, these methods often lack the depth needed to foster the level of self-reflection required for meaningful stress assessments, sometimes yielding superficial or mechanical responses [5]. Professional counseling, on the other hand, facilitates deeper self-reflection but is typically expensive, time-consuming, and less accessible to a broad student population [6]. These limitations highlight the need for innovative, accessible tools that promote in-depth self-reflection, a crucial process for achieving more meaningful stress assessments.

To address these challenges, chatbot-based assessment tools have emerged as promising alternatives, combining accessibility with the potential to promote self-reflective conversation in a supportive environment [7, 8]. However, despite their inherent capabilities, many existing chatbot-based assessments merely replicate the format of traditional self-report questionnaires, failing to fully leverage the interactive and adaptive features that chatbots can offer [9, 10]. This limitation highlights the need for chatbots that move beyond simple

question-and-answer formats, actively encouraging deeper reflection to elicit richer, more authentic responses, ultimately enabling more precise and meaningful assessments of academic stress [11].

One such approach involves leveraging self-disclosure, where the chatbot shares its own thoughts or emotions to establish a more engaging and empathetic exchange. Research suggests that self-disclosure by the chatbot can encourage users to share their experiences more openly, thereby promoting deeper self-reflection [12, 13]. Despite its promise, there remains a significant gap in the development of self-disclosing chatbots specifically designed to support meaningful academic stress assessment, along with limited guidelines on effectively designing such chatbots in educational settings.

In this study, we address this gap by developing and comparing two chatbot conditions: the non-self-disclosure (NSD) chatbot and the self-disclosure (SD) chatbot. We analyzed semi-structured interviews with participants to investigate (1) how chatbot features influence student self-reflection, (2) the impact of chatbot interactions on students' completion of self-report questionnaires, and (3) students' expectations and concerns regarding the use of self-disclosure chatbots in educational settings. The findings provide valuable insights into designing chatbots that foster in-depth self-reflection, supporting meaningful academic stress assessment and management in higher education.

2. Related work

2.1 Chatbot-based stress assessments

With the development of large language models (LLMs) like ChatGPT, interest has surged in using chatbots for stress assessment, offering a promising alternative to conventional methods such as self-report questionnaires and professional counseling [14]. LLM-powered chatbots provide intuitive conversational interfaces that not only improve accessibility but also enhance user engagement, laying the groundwork for more effective and responsive chatbot-based assessment strategies [15].

Typically, chatbots are programmed to digitize standard questionnaires into conversational formats, increasing accessibility but often lacking in fostering genuine engagement. For instance, Moldt et al. [9] integrated the Perceived Stress Questionnaire into a chatbot but observed that this approach did little to encourage in-depth, reflective dialogue. Similarly, Toulme et al. [10] found that their depression-screening chatbot, while comparable to online questionnaires in efficiency, lacked the natural conversational flow necessary to make users feel at ease, thereby diminishing the quality of the assessment.

Despite the advancements in chatbot technology, existing chatbot-based stress assessments often fail to promote the profound self-reflection essential for meaningful assessments. Recognizing this limitation, our study aims to design chatbots that go beyond merely digitizing and automating assessments. Instead, we focus on actively facilitating deeper self-reflection, which not only helps students engage in more in-depth conversations but also leads to more precise and meaningful assessments of their academic stress.

2.2 Self-disclosure in chatbots

Recent studies highlight the potential of chatbot self-disclosure in promoting users' self-reflection [16]. When chatbots share thoughts or emotions in response to user input, they create a sense of trust and reflective engagement that is essential for meaningful stress assessment [17]. Through thoughtfully designed self-disclosure interactions, students can be prompted to reflect more deeply on their individual stressors, symptoms, and coping strategies, leading to a more profound understanding of their academic stress.

Barak and Gluck-Ofri [12] identified three core elements of self-disclosure—information, thoughts, and feelings—each with varying levels of intensity. Research by Lee et al. [18] found that chatbots utilizing high levels of self-disclosure tend to encourage users to reciprocate, sharing more meaningful and personal insights. Similarly, Croes et al. [13] demonstrated that self-disclosing chatbots increase user trust, empathy, and emotional connection, while Park et al. [19] showed

that these chatbots enhance user engagement and satisfaction. By creating an environment that encourages open self-disclosure, chatbots can facilitate the kind of self-reflection that enables students to gain clearer insights into their academic stress [20].

Despite these findings, most research on self-disclosure in chatbots has centered on improving general user experience rather than its specific role in fostering self-reflection for academic stress assessment. Our study aims to address this gap by investigating how self-disclosure within chatbots can promote deeper self-reflection among students, particularly in the context of academic stress assessment.

3. Materials and methods

3.1 Participants

We recruited 20 university students, all engineering majors with at least one year of higher education, to participate in this study. Participants were randomly assigned to one of two experimental conditions: the non-self-disclosure (NSD) chatbot condition ($n = 10$) and the self-disclosure (SD) chatbot condition ($n = 10$). Each condition included an equal distribution of five male and five female participants.

The average age of participants in the NSD chatbot condition was 23.1 ± 1.52 years, while those in the SD chatbot condition had an average age of 22.9 ± 1.45 years. Statistical analysis revealed no significant differences in age distribution between the two conditions ($t = 0.3$, $p = 0.77$). All recruitment procedures and study protocols were approved by the Institutional Review Board at Seoul National University of Science and Technology (IRB approval number: 2023-0021-01).

3.2 Academic stress assessment questionnaire

In this study, the SISCO-AS [4] was utilized to design chatbots that support self-reflection on academic stress and was subsequently employed to assess academic stress. The SISCO-AS consists of 31 items that assess various dimensions of academic stress, including stressors, symptoms, and coping strategies. The questionnaire begins with a yes/no qualifier item, followed by a question

assessing the overall intensity of academic stress. The remaining 29 items are divided into three sections: eight items measure exposure to specific stressors, 15 assess the intensity of symptoms, and six evaluate the use of coping strategies. Each item is rated on a 5-point Likert scale, ranging from 1 ("never") to 5 ("always").

3.3 Two different chatbot-based stress assessment conditions

To investigate the effect of self-disclosure in chatbots on student self-reflection related to academic stress, we developed two chatbot conditions based on the SISCO-AS: the NSD chatbot and the SD chatbot. Both chatbots were created using the GPT-4o API through prompt engineering and implemented in Streamlit, allowing participants to interact via links on their laptops. Each chatbot adopted the persona of a friendly and empathetic peer to create a relatable and comfortable interaction.

The interactions were structured into four sessions: (1) small talk, (2) self-disclosure (SD condition only), (3) key questions, and (4) closing. In the NSD condition, the chatbot maintained a neutral, task-focused tone, avoiding any sharing of personal experiences. In contrast, the SD chatbot shared personal thoughts and emotions to encourage students' self-reflection, aligning with the self-disclosure principles outlined by Barak and Gluck-Ofri [12].

Both chatbots initiated with a friendly greeting in the (1) small talk session. Following this, the NSD chatbot transitioned directly to the (3) key questions session, while the SD chatbot engaged in a (2) self-disclosure session, sharing relevant thoughts or emotions to foster students' self-reflection (see Figure 1). In the (3) key questions session, both chatbots asked open-ended questions to explore participants' stressors, symptoms, and coping strategies, structured according to the SISCO-AS questionnaire. The SD chatbot periodically interwove relevant personal experiences by alternating between (2) self disclosure and (3) key questions session, enhancing the conversational flow and prompting students to reflect more deeply on their experiences. In the (4)

closing session, both chatbots expressed appreciation, with the NSD chatbot concluding simply with thanks, while the SD chatbot added a personal message.

3.4 Experimental procedures

The experiment was conducted in a quiet, private setting to ensure participants could focus fully on each task. Participants first engaged in the chatbot interaction, followed by completing the SISCO-AS questionnaire. Although participants were informed they could terminate the experiment at any point if they felt uncomfortable, all participants chose to complete the experiment. Following these activities, each participant took part in a semi-structured interview aimed at exploring how chatbot features shaped participants' self-reflection, impacted their experience with the SISCO-AS questionnaire, and provided insights into their perspectives on incorporating self-disclosure chatbots into educational settings. The entire session lasted approximately 42.22 ± 6.7 minutes, with participants spending 12.39 ± 4.29 minutes on the chatbot interaction, around five minutes on the SISCO-AS questionnaire, and about 30 minutes in the interview.

3.5 Data analysis

The academic stress levels of participants were assessed using the SISCO-AS. To determine whether there were significant differences of academic stress levels between the NSD and SD chatbot conditions, we conducted an independent samples t-test on the SISCO-AS percentile scores.

For the qualitative analysis, the semi-structured interview responses were analyzed using thematic analysis. First author initially coded the interview responses to identify emerging themes, and the rest of authors independently reviewed and refined these codes. Through an iterative process of discussion and revision, the final themes were refined to ensure consistency, and alignment with the study's research objectives. This process led to the identification of six primary themes.

4. Results

4.1 Academic stress levels of participants

Before analyzing how chatbot interactions

influenced self-reflection, we verified whether there were any pre-existing differences in academic stress levels between the two conditions. Participants' academic stress levels were assessed using percentile scores derived from the SISCO-AS, which they completed after their assigned chatbot interaction. Statistical analysis showed no significant difference between the NSD chatbot condition (47.48 ± 10.29) and the SD chatbot condition (45.38 ± 12.79), $t = 0.77$, $p = 0.45$, indicating that the two conditions were comparable in terms of academic stress baseline measures.

4.2 Influence of chatbot features on student self-reflection

This section identifies two of the six primary themes that emerged regarding how chatbot features—*reflective inquiry* and *perceived agency in the conversation*—influenced the self-reflection experiences of participants in the NSD and SD chatbot conditions. Responses are labeled as either from the NSD chatbot condition ('NSD') or the SD chatbot condition ('SD').

Reflective inquiry

Reflective inquiry, which depends on how effectively each chatbot facilitated reflective questions, was a key factor in determining the depth of self-reflection across the both chatbot conditions. In the NSD chatbot condition, participants generally felt that the chatbot's responses were mechanical and superficial, which hindered their ability to engage in self-reflection. One participant expressed frustration, saying, "The chatbot kept asking the same kinds of questions, and they felt more like a survey than a conversation. I didn't feel like it cared about my responses" (NSD #7). Another participant noted, "It didn't make me think deeply at all. I just answered quickly without feeling a connection" (NSD #5). The lack of empathy in the chatbot's questions also contributed to a sense of detachment, as one participant shared, "It didn't feel like the chatbot was listening, so I didn't bother reflecting deeply" (NSD #10). These responses indicate that the NSD chatbot's limited ability to prompt reflective inquiry, combined with its neutral tone, stifled opportunities for deeper self-reflection.

In contrast, the SD chatbot was perceived as much more effective in prompting reflective inquiry, primarily due to the emotional engagement it brought into the conversation. Participants in the SD chatbot condition noted that the chatbot's empathetic tone encouraged them to think more critically about their academic stress. One participant shared, "The chatbot asked thoughtful questions that really made me consider things I hadn't before, and its empathy made me feel comfortable enough to dig deeper into my thoughts" (SD #7). Another participant emphasized how the chatbot's shared experiences influenced their reflection: "When the chatbot told its own stories, it made me reflect on my own situations, and that led to a much deeper understanding of my stress" (SD #2). The combination of reflective questioning and empathetic responses created an environment where participants felt supported and could engage in in-depth self-reflection.

Perceived agency in the conversation

Perceived agency, or the sense of control participants felt over the conversation's direction and flow, was another critical factor influencing self-reflection. In the NSD chatbot condition, participants frequently felt restricted by the chatbot's rigid structure, limiting their ability to steer the conversation in meaningful ways. One participant expressed, "It felt like I was just responding to the chatbot's prompts without much room to share what I really wanted to talk about" (NSD #5). Another participant added, "The chatbot didn't seem to care about what I said—it just kept asking questions, so I didn't feel in control of the conversation at all" (NSD #4). This lack of agency led to a more passive experience, where participants felt they were being guided through a predetermined script rather than engaging in a dynamic, two-way interaction. This diminished sense of agency made it difficult for participants to explore their thoughts and feelings deeply, as they felt constrained by the chatbot's predefined structure.

In contrast, participants in the SD chatbot condition reported a stronger sense of agency, which empowered them to explore their thoughts more freely. One participant noted, "The chatbot gave me

space to talk about what was important to me, and I felt like I was leading the conversation at times" (SD #4). This conversational flexibility allowed participants to steer the dialogue toward topics that mattered most to them, fostering deeper self-reflection. Another participant commented, "I felt like I had control over where the conversation was going, and that helped me reflect on my stress in a way that felt more personal" (SD #10). Participants also appreciated the balance between the chatbot's guidance and their own input: "I felt like I had the freedom to talk about what I needed to, and the chatbot would follow along and engage in a deep conversation with me about it" (SD #1). This sense of agency enhanced participants' engagement, as they felt more invested in the conversation and, as a result, more willing to reflect deeply on their academic stress.

4.3 Impact of chatbot interaction on completion of self-report questionnaires

Participants' interactions with the chatbots distinctly shaped the depth of participants' reflections and the quality of their responses to the self-report questionnaire. Two primary themes emerged from the analysis, illustrating how the features of the NSD and SD chatbots influenced participants' approaches to the self-report questionnaire: *clarity in identifying stress factors* and *planning for future actions*.

Clarity in identifying stress factors

The degree to which participants were able to clearly identify and articulate their stress factors in the self-report questionnaire varied significantly between the NSD and SD chatbot conditions. In the NSD chatbot condition, many participants struggled to pinpoint specific stressors even after interacting with the chatbot. The NSD chatbot's structured and less interactive nature seemed to limit participants' ability to connect the conversation with the stress factors they were asked to identify. As one participant explained, "I found it hard to connect the chatbot conversation to specific questions in the questionnaire, as I didn't feel like my stress was any clearer" (NSD #2). Another participant noted, "The questions on the form felt disconnected from what I talked about with the chatbot" (NSD #9). This

disconnect between the chatbot conversation and the questionnaire content hindered participants from gaining the clarity needed to articulate their stressors effectively.

In contrast, participants in the SD chatbot condition reported greater clarity when answering the self-report questionnaire. The SD chatbot's personalized and engaging prompts helped many participants identify their stress factors more confidently. As one participant reflected, "After talking with the chatbot, I could see my stress more clearly and answer the questionnaire in a more focused way" (SD #6). Another participant added, "I felt like the chatbot had already helped me break down my stress, which made the questionnaire easier to fill out" (SD #7). These experiences suggest that the SD chatbot's empathetic and interactive nature played a crucial role in helping participants articulate their stress factors with greater precision.

Planning for future actions

The NSD and SD chatbot interactions shaped participants' experience of completing the questionnaire in distinct ways, especially regarding the future planning. In the NSD chatbot condition, participants found the interaction useful for organizing their thoughts, but it rarely led to actionable future plans. One participant noted, "Answering the chatbot's questions and the questionnaire somewhat helped me summarize my thoughts on stress, but it didn't make me think about what to do next" (NSD #7). Another echoed this, saying, "Interacting with the chatbot and checking the self-report questionnaire made me aware of my academic stress, but I didn't come away with new ideas on how to handle it" (NSD #1). Although the NSD chatbot allowed for some awareness, participants felt it lacked the depth required to inspire future coping strategies. One participant remarked, "It would have been more helpful if the chatbot guided me on what actions to take when academic stress arises" (NSD #8). Ultimately, completing the questionnaire after the NSD chatbot interaction didn't lead to deeper insights or concrete plans for managing stress, likely due to the chatbot's surface-level prompts.

On the other hand, participants in the SD chatbot

condition often reported that the interaction not only led them to reflect on their stress but also motivated them to plan for future actions. One participant said, "The chatbot helped me think about my current coping methods and how I can use them more effectively in the future" (SD #4). Another mentioned, "Thanks to the chatbot's empathy, I reflected deeply and realized that I should apply similar strategies in future stressful situations" (SD #10). This proactive approach was common in the SD condition, where participants felt empowered to take specific steps moving forward. As one participant put it, "Filling out the questionnaire after interacting with the chatbot made me recognize new strategies for managing stress and inspired me to try them in the future" (SD #8).

4.4 Expectations and concerns regarding self-disclosure chatbots in education

The integration of self-disclosure chatbots into educational settings elicits both considerable expectations and significant concerns. While participants highlighted their enthusiasm for the potential advantages of *personalized support* offered by such chatbots, they also raised important concerns, particularly around *privacy and data security*.

Personalized support

Participants expected the chatbot to provide personalized and timely emotional support, with many appreciating its ability to offer continuous, individualized feedback based on their unique stressors. One participant noted that having such chatbot available whenever needed will be helpful for immediate stress relief, something not always possible with traditional counseling resources (SD #6). Others expressed a desire for the chatbot to learn from previous interactions and offer progressively more tailored advice, with one suggesting that this feature would make the chatbot feel more like a trusted and adaptive mentor (NSD #4). Overall, participants saw chatbots as a tool to enhance self-awareness, provide real-time support, and improve stress management through continuous engagement.

Privacy and data security

While participants recognized the potential benefits of chatbots, they expressed significant concerns about privacy and data security. Some were uneasy about sharing sensitive information due to fears of data breaches or unauthorized access, with one participant noting the risk of personal data being exposed (NSD #5). Concerns also arose around data usage, as participants wanted transparency about who could access their data and how it was stored (SD #3). Additionally, some participants were worried about their ability to control their data, such as having the option to delete personal information if needed (NSD #8). These concerns highlighted the need for clearer data management practices to build trust in chatbot technology.

5. Discussion and conclusion

This study explores the impact of self-disclosure in chatbots for enhancing students' self-reflection in academic stress assessments. By analyzing responses from both questionnaire and interviews, the findings reveal that the SD chatbot facilitated deeper conversations, helping participants more clearly identify their stressors and plan future coping strategies. In contrast, the NSD chatbot was less engaging, leading to lower levels of interpersonal depth to elicit profound reflection.

A plausible explanation for these results lies in the SD chatbot's ability to foster an empathetic environment through self-disclosure and contextually relevant inquiries. The SD chatbot encouraged participants to move beyond basic question-and-answer exchanges, enabling deeper self-reflection on their thoughts and feelings. These findings align with prior research highlighting the importance of reciprocity in self-disclosure, where chatbot-initiated sharing and adaptive responses prompt users to share deeper reflections on their own experiences [18, 19]. The SD chatbot's integration of conversational empathy and strategically designed prompts demonstrates how such features can effectively support students in clarifying their stressors and developing actionable coping strategies.

5.1 Implications and design recommendations

Based on our findings, we propose several practical recommendations for designing chatbots that

support self-reflection and yield more meaningful academic stress assessments. First, incorporating empathetic self-disclosure—through personal or hypothetical stories—helps establish a reciprocal relationship with students, encouraging them to share more openly and engage in deeper self-reflection. Second, chatbots should use reflective inquiry by asking thoughtful, open-ended questions that prompt students to examine their experiences, avoiding repetitive or formulaic prompts. Third, conversational flexibility is essential; allowing students to steer the direction of the conversation enhances personalization and engagement, fostering more in-depth self-reflection. Finally, prioritizing privacy and data security is critical. Addressing students' concerns about data handling and implementing robust security measures builds trust, thereby encouraging open self-disclosure.

5.2 Limitations and future directions

Building on the findings, this study has several limitations that should be addressed. The small sample size and focus on engineering majors may limit the generalizability of the findings, indicating a need for larger, more diverse samples across academic disciplines to validate these results. Future research should examine chatbot effectiveness across a broader demographic, including students from various academic backgrounds and cultural contexts. Additionally, while the SD chatbot appeared to promote self-reflection, this study did not include objective measures to determine whether it led to deeper self-reflection and more meaningful academic stress assessments.

Despite these limitations, this study provides critical insights into the potential of self-disclosing chatbots to enhance self-reflection for meaningful academic stress assessments. By fostering deeper engagement and enabling clearer identification of stressors, self-disclosing chatbots demonstrate their value as tools for supporting students' academic well-being. These findings lay the groundwork for developing empathetic and effective chatbot-based assessment systems that can help students better understand and manage their academic stress, ultimately contributing to improved mental health outcomes in educational

settings.

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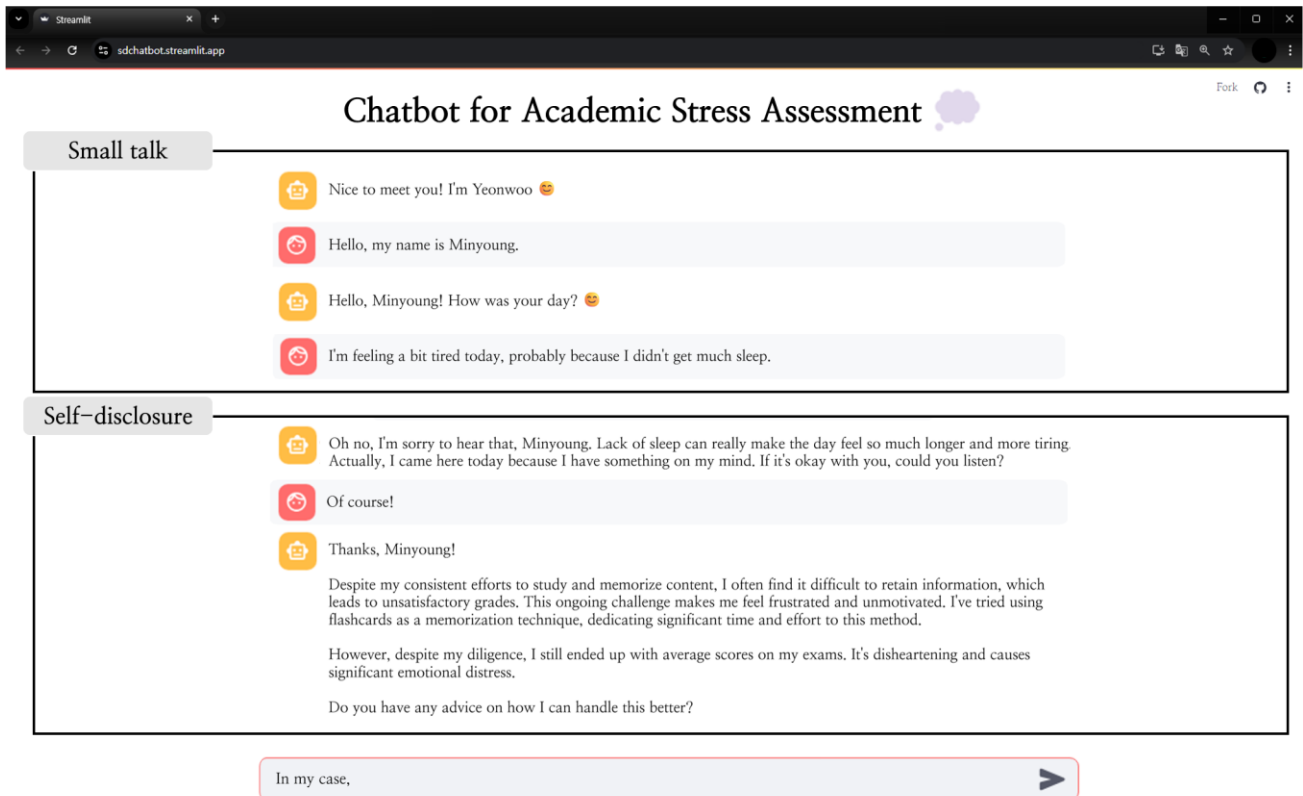


Figure 1. Example of student interaction with the self-disclosure (SD) chatbot: small talk and self-disclosure sessions